Scenario 4B: Low and heterogeneous exposure stress test

MSSTNB posterior performance, exposure-group recovery, and oracle diagnostics

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1 Purpose and main conclusion

Scenario 4B is designed as a **low and heterogeneous exposure stress test** for the MSSTNB model. It is not intended to duplicate Scenario 4A. Scenario 4A lowered the observation-level intensity and created a global sparse-count regime. Scenario 4B instead lowers and heterogenizes the known exposure term while preserving the underlying temporal-spatial latent structure.

The main conclusion is:

> Low and heterogeneous exposure creates local information imbalance. In stable fits, the main consequence is weaker latent-risk recovery in low-exposure areas than in high-exposure areas. In difficult replicates, the same weak-information geometry can trigger posterior ridges among regression effects, spatial effects, and latent temporal risk. Oracle diagnostics confirm that these failures are not due to a generic beta/r/kappa coding error.

The key evidence is:

1. S4B has higher overall counts than S4A but still has a 20% numerical-instability rate.
2. Among stable and soft-warning fits, β_1 remains stable, whereas β_2 remains fragile across replicates.
3. Low-exposure areas have much worse log-λ recovery than high-exposure areas.
4. Oracle-λ diagnostics restore β_2 but leave β_0, ϕ, and κ unstable.
5. Oracle-$\lambda + \phi$ diagnostics restore all regression effects and eliminate numerical guards.

2 Scenario design

2.1 What is changed in Scenario 4B?

Scenario 4B uses the same reference dynamic structure as Scenario 3, but changes the known exposure. The S4B exposure is constructed as

$$e_{tj}^{\mathrm{S4B}} = e_{tj}^{\mathrm{ref}} m_{tj},$$

where m_{tj} is a low and heterogeneous exposure multiplier. The official setting uses an average multiplier close to 0.05, so the average exposure is approximately 5% of the reference exposure. The multipliers are heterogeneous across areas and mildly variable over time.

The official S4B data design is:

Table 1: Official S4B low-exposure data-generation design.

quantity	value
Scenario ID	S4B_LOW_HETEROGENEOUS_EXPOSURE_T100
Stress source	low and heterogeneous exposure
Target mean multiplier	0.05
Area heterogeneity	area_log_sd = 0.75
Time heterogeneity	time_log_sd = 0.08
Lower multiplier bound	0.005
Upper multiplier bound	0.25
T	100
n1	9

reps	10

This design makes the overall count level lower than the S3 reference setting but not as globally sparse as S4A. The stress is therefore not only the total number of observed cases; it is the **uneven distribution of information across exposure-poor and exposure-rich areas**.

2.2 Difference from S4A

The distinction between S4A and S4B is essential:

- S4A asks what happens when the whole observation process becomes sparse.
- S4B asks what happens when exposure is low and heterogeneous, so some areas have much weaker information than others even though the overall dataset still contains moderate counts.

Thus, S4B is closer to a small-area exposure-imbalance stress test.

3 Data calibration

The S4B data-generation check passed all calibration criteria. The realized average exposure multiplier is essentially the target value of 0.05, while the exposure coefficient of variation is large enough to create meaningful exposure heterogeneity.

Table 2: S4B data-calibration summary across 10 replicates.

metric	value
Mean exposure ratio	0.050
Mean count	9.558
Median count	5.100
Zero proportion	0.125
Total count	8.60e+04
Exposure CV	0.825
Low-exp mean count	3.441
High-exp mean count	17.647
Low-exp zero prop	0.214
High-exp zero prop	0.067

The low-exposure group has average count about 3.44 and zero proportion about 0.214, while the high-exposure group has average count about 17.65 and zero proportion about 0.067. This confirms that S4B creates a within-dataset information contrast: some areas are close to a sparse regime, while others remain relatively informative.

4 Fitting strategy

S4B is fitted using the same dynamic model structure as S3 and the same fixed-gamma stabilized machinery used in S4A diagnostics.

The fitted model estimates:

$$\beta_0, \beta_1, \beta_2, \phi, \tau_\phi, r, \kappa, \lambda.$$

The model fixes:

$$\gamma = 0.8.$$

The model disables δ and ω updates. This fixed-gamma strategy is intentional: S4B is designed to isolate exposure stress, not to test weak identification of γ.

As in S4A, numerical guards are recorded and used as diagnostics. The purpose is not to hide difficult replicates, but to classify them as stable, soft warning, or numerical instability.

5 Fit-status classification

Each replicate is classified into one of three groups:

- **Stable:** no meaningful numerical guards.
- **Soft warning:** small guard activity without severe parameter or latent-process failure.
- **Numerical instability:** large guard activity, extreme β_0, extreme λ RMSE, or extreme log-λ RMSE.

Table 3: S4B fit-status counts.

status	n_reps	prop
Unstable	2	20%
Soft warning	1	10%
Stable	7	70%

The result is 7 stable fits, 1 soft warning, and 2 numerical-instability replicates. Thus, S4B should be interpreted as an exposure stress test with a non-negligible failure rate, not as a setting where all replicates are ordinary successful posterior fits.

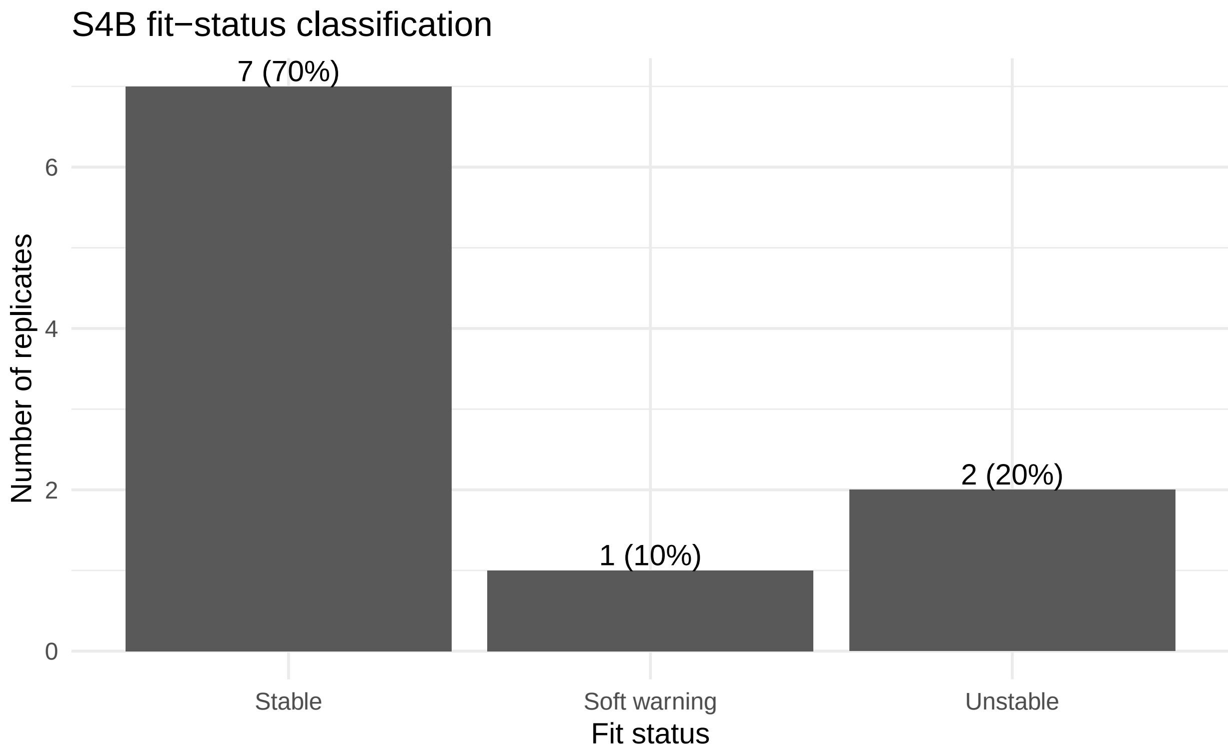

Figure 1: S4B fit-status classification across 10 replicates.

5.1 Replicate-level summaries

The replicate-level results are split into narrow tables to avoid over-wide PDF tables.

Table 4: Replicate-level S4B data and exposure summaries.

rep	status	mean	zero	exp_ratio	exp_cv
1	Stable	9.01	0.116	0.051	0.773
2	Soft warning	11.90	0.153	0.046	0.785
3	Stable	8.40	0.130	0.056	1.185
4	Unstable	5.90	0.188	0.055	0.854
5	Stable	15.78	0.113	0.049	0.850
6	Stable	9.36	0.072	0.049	0.648
7	Unstable	7.09	0.166	0.047	0.823
8	Stable	7.54	0.191	0.047	0.801
9	Stable	9.33	0.084	0.051	0.873
10	Stable	11.26	0.039	0.042	0.657

Table 5: Replicate-level S4B regression and global lambda recovery summaries.

rep	status	beta0	beta1	beta2	loglam_rmse	cor_loglam
1	Stable	-1.708	0.500	-0.773	0.203	0.022
2	Soft warning	-1.563	0.497	-1.425	0.265	0.383
3	Stable	-1.803	0.520	2.666	0.172	0.200
4	Unstable	-2.19e+04	0.586	-0.815	3.009	0.733
5	Stable	-1.497	0.472	-2.258	0.294	0.547

rep	status	beta0	beta1	beta2	loglam_rmse	cor_loglam
6	Stable	-1.766	0.497	1.941	0.137	0.407
7	Unstable	-2.899	0.502	-4.894	54.310	0.652
8	Stable	-1.866	0.491	-1.883	0.197	0.308
9	Stable	-1.506	0.507	-0.763	0.117	0.535
10	Stable	-1.308	0.482	-1.335	0.092	0.498

Table 6: Replicate-level S4B exposure-specific lambda recovery and numerical guards.

rep	status	low_loglam	high_loglam	beta_guard	kappa_guard	lambda_guard
1	Stable	0.156	0.077	0	0	0
2	Soft warning	0.397	0.076	0	0	3
3	Stable	0.245	0.124	0	0	0
4	Unstable	0.113	3.794	39798	3631860	1053548
5	Stable	0.450	0.040	0	0	0
6	Stable	0.175	0.079	0	0	0
7	Unstable	0.153	96.752	1067	562	817650
8	Stable	0.225	0.075	0	0	0
9	Stable	0.153	0.100	0	0	0
10	Stable	0.114	0.080	0	0	0

Replicates 4 and 7 are the numerical-instability replicates. Replicate 4 shows a severely negative posterior mean for β_0 and large guard activity. Replicate 7 has catastrophic lambda RMSE and substantial lambda-output guard activity. These two replicates should not be averaged with stable fits as ordinary posterior-recovery results.

6 Posterior performance by subset

Table 7: S4B data and exposure summaries by fit-status subset.

subset	n	mean	zero	exp_ratio	exp_cv	low_count	high_count
All	10	9.56	0.125	0.049	0.825	3.44	17.65
Stable	7	10.10	0.107	0.049	0.827	3.44	19.66
Stable+soft	8	10.32	0.112	0.049	0.822	3.20	19.83
Unstable	2	6.50	0.177	0.051	0.839	4.40	8.93

Table 8: S4B regression recovery by fit-status subset.

subset	n	beta1_mean	beta1_sd	beta1_cov	beta2_mean	beta2_sd	beta2_cov
All	10	0.505	0.031	0.900	-0.954	2.109	0.400
Stable	7	0.496	0.016	1.000	-0.344	1.900	0.429
Stable+soft	8	0.496	0.015	1.000	-0.479	1.800	0.500
Unstable	2	0.544	0.059	0.500	-2.855	2.884	0.000

Among stable-plus-soft-warning fits, β_1 remains accurately recoverable, with average posterior mean about 0.496 and coverage equal to 1.0. In contrast, β_2 has average posterior mean close to the truth, about -0.479, but its between-replicate standard deviation is about 1.800 and its coverage is only about 0.5. This indicates coefficient-specific fragility: exposure stress does not destroy all regression information, but it makes the β_2 direction weakly identified.

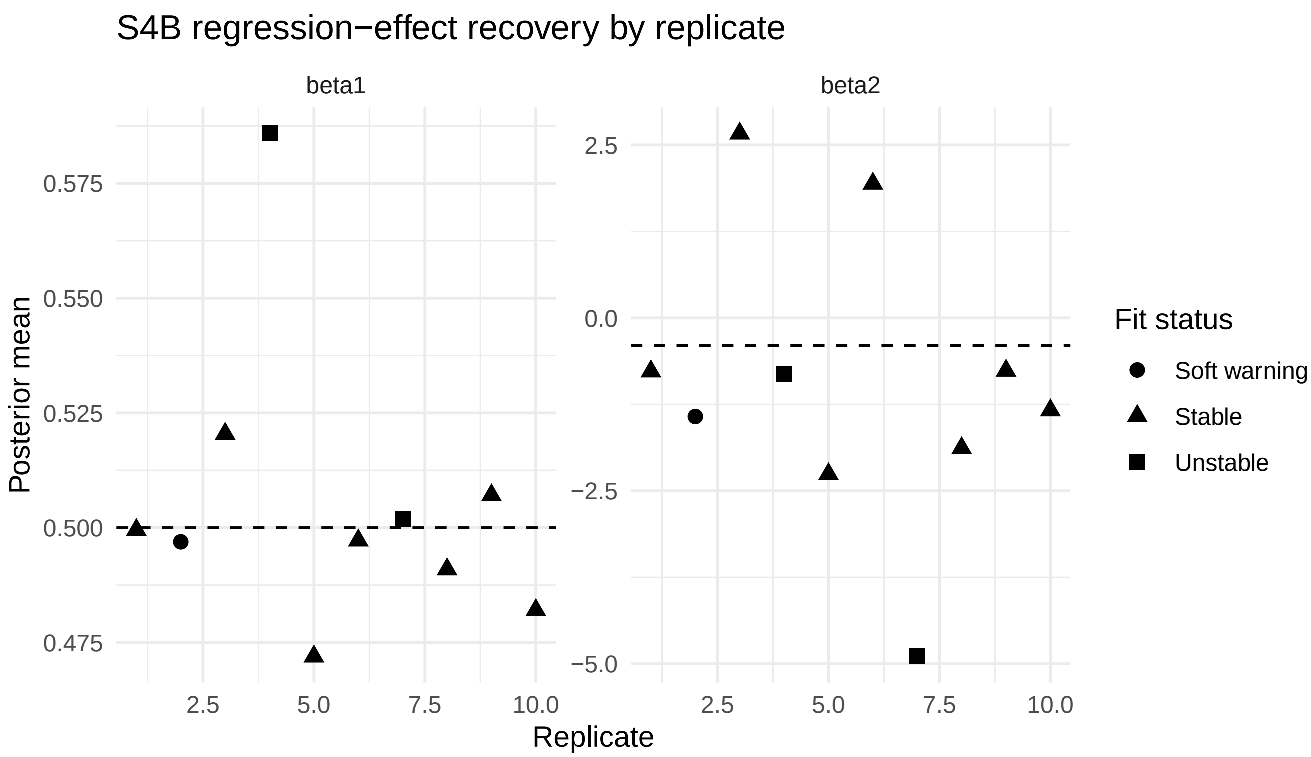

Figure 2: Posterior means for beta1 and beta2 by replicate. Dashed lines show the truth.

7 Low- versus high-exposure recovery

The central S4B result is the low- versus high-exposure contrast.

Table 9: Low- versus high-exposure count and log-lambda recovery contrasts.

subset	n	low_count	high_count	low_zero	high_zero	low_loglam	high_loglam	diff
All	10	3.44	17.65	0.214	0.067	0.218	10.120	-9.902
Stable	7	3.44	19.66	0.199	0.010	0.217	0.082	0.135
Stable+soft	8	3.20	19.83	0.230	0.010	0.239	0.081	0.158
Unstable	2	4.40	8.93	0.150	0.295	0.133	50.273	-50.140

In the stable-only subset, low-exposure areas have average count about 3.44 and zero proportion about 0.199, while high-exposure areas have average count about 19.66 and zero proportion about 0.010. The corresponding log-λ RMSE is about 0.217 in low-exposure areas but only about 0.082 in high-exposure areas.

In the stable-plus-soft-warning subset, the contrast is similar: low-exposure log-λ RMSE is about 0.239, while high-exposure log-λ RMSE is about 0.081.

Therefore, even when the full MCMC fit is numerically stable, local latent temporal-risk recovery is substantially weaker in low-exposure areas.

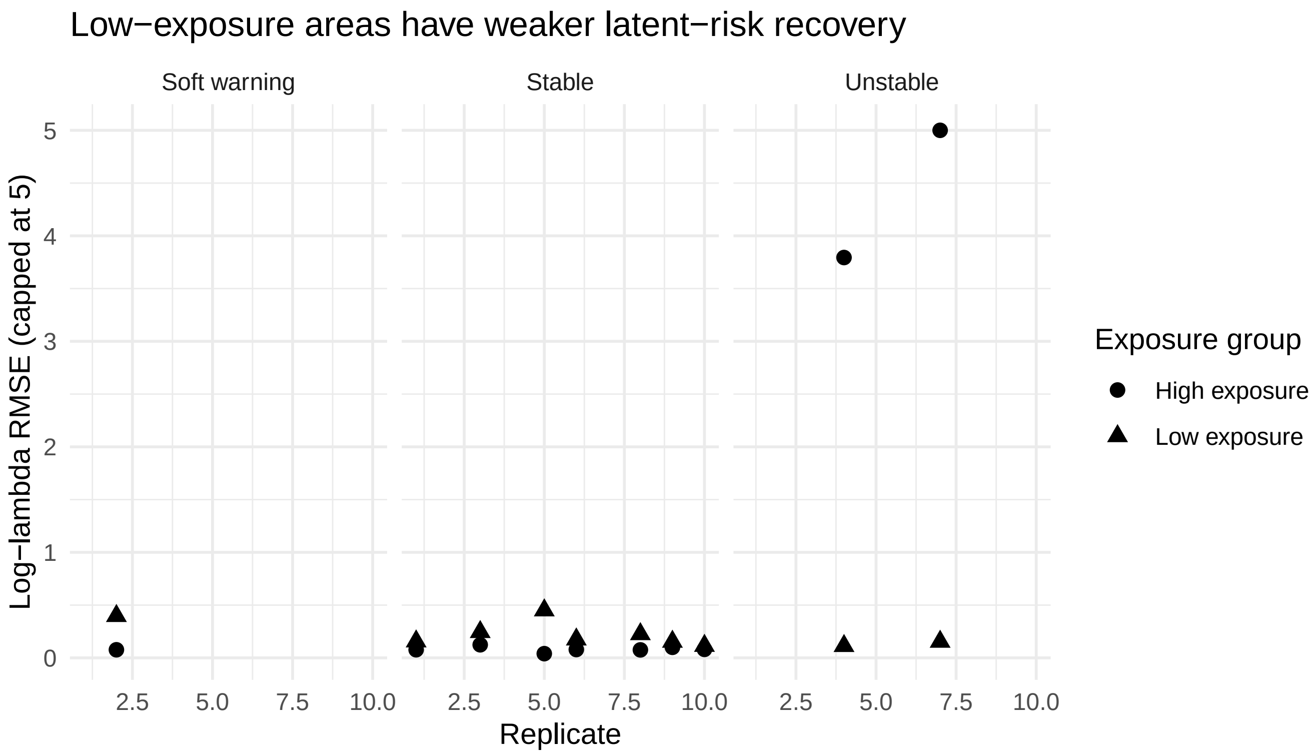

Figure 3: Low- versus high-exposure log-lambda RMSE by replicate. Values are capped at 5 for display.

8 Comparison with S3 and S4A

Table 10: Comparison of S3, S4A, and S4B summaries.

scenario	subset	n	mean	zero	beta2_mean	beta2_sd	loglam_rmse	cor_loglam	instab
S3 control	all_control_reps	3	195.05	0.000	-0.408	0.009	0.047	0.734	0.00
S4A sparse	Stable+soft	8	3.17	0.223	-0.675	2.528	0.206	0.435	0.20
S4B exposure	Stable+soft	8	10.32	0.112	-0.479	1.800	0.185	0.363	0.20

This comparison shows that S4B is less globally sparse than S4A: the stable-plus-soft-warning S4B mean count is about 10.32 and zero proportion is about 0.112, compared with S4A mean count about 3.17 and zero proportion about 0.223. However, S4B still has degraded β_2 recovery, degraded log-λ recovery, and a 20% numerical-instability rate. This supports the interpretation that exposure heterogeneity can expose similar identifiability boundaries even when global sparsity is milder.

9 Diagnostic ladder

The diagnostic ladder combines the stable-plus-soft-warning sampled-lambda S4B summary, the S3 and S4A references, and the two oracle diagnostics. The two S4B numerical-instability replicates are reported separately through the fit-status summary and are not averaged into the main sampled-lambda recovery row.

Table 11: S4B diagnostic ladder: numeric summary.

step	n	beta1	beta2	beta2_sd	loglam_rmse	cor_loglam	kappa_guard
S4B stable+soft	8	0.496	-0.479	1.800	0.185	0.363	0
S3 control	3	0.492	-0.408	0.009	0.047	0.734	0
S4A sparse	8	0.511	-0.675	2.528	0.206	0.435	NA
Oracle lambda	2	0.513	-0.458	0.037	0.000	1.000	4.67e+05
Oracle lam+phi	2	0.513	-0.389	0.046	0.000	1.000	0

Table 12: S4B diagnostic ladder: interpretation.

step	conclusion
S4B stable+soft	Main exposure-stress fit summarized on stable+soft reps; failure rate is reported separately.
S3 control	Same machinery stable in non-exposure-stressed reference setting.
S4A sparse	Global sparse-count stress; useful reference for S4B but not the same stress source.
Oracle lambda	Fixing lambda restores beta2, but beta0/phi/kappa can remain unstable.
Oracle lam+phi	Fixing both lambda and phi restores beta/r and eliminates numerical guards.

The ladder has the same logic as S4A. The S3 control confirms that the same stabilized machinery can work in a non-stressed reference setting. The S4B stable-plus-soft-warning row summarizes conditional recovery among nonfailed fits, while the fit-status table reports the 20% instability rate. The oracle-λ diagnostic tests whether the β_2 problem is tied to latent temporal-risk confounding. The oracle-$\lambda + \phi$ diagnostic tests whether the beta/r block is itself valid once latent temporal and spatial effects are fixed.

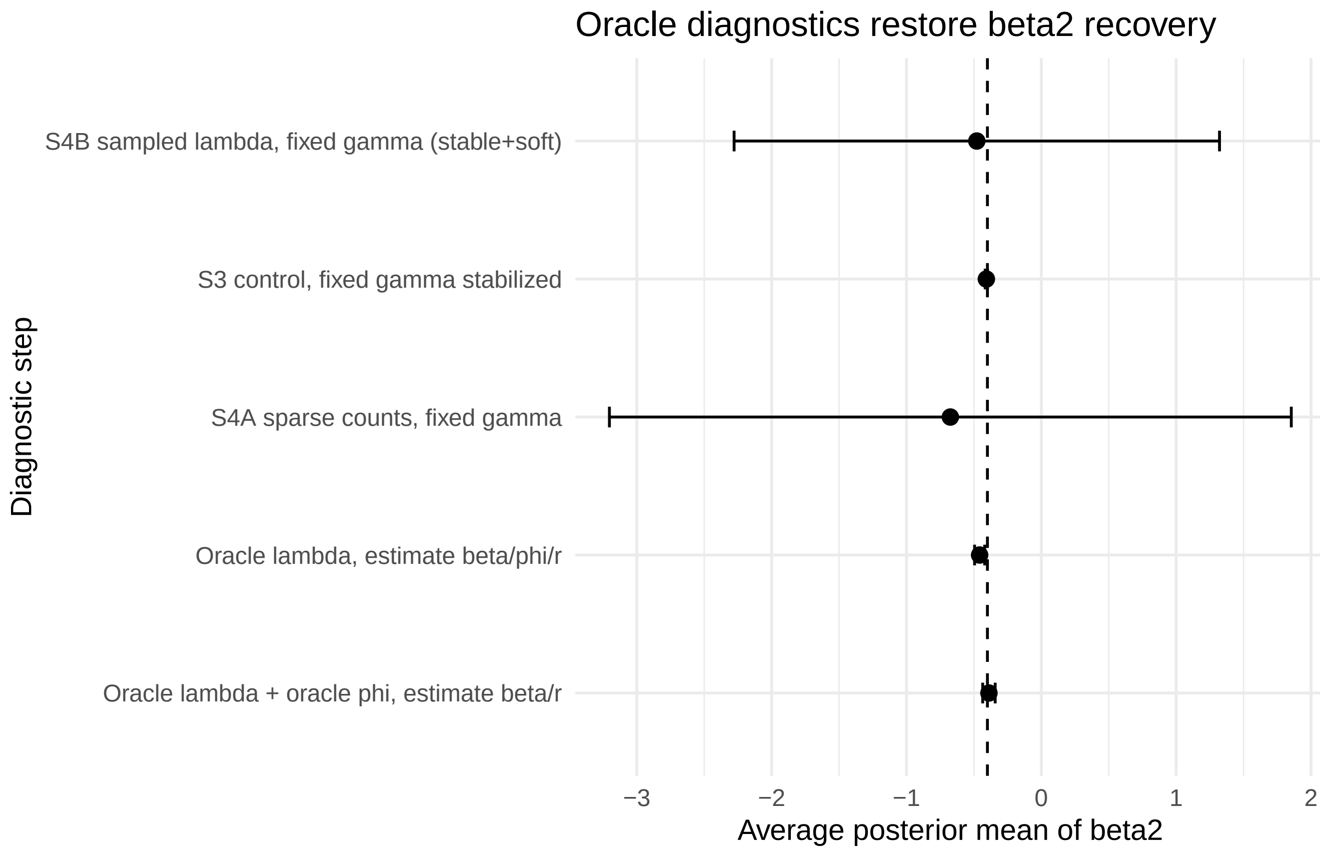

Figure 4: Beta2 recovery across the S4B diagnostic ladder. The dashed line shows the truth.

10 Oracle diagnostics in detail

The oracle diagnostics were run for the two numerical-instability replicates, reps 4 and 7.

10.1 Step 1: Oracle lambda, estimate beta/phi/r

In this diagnostic, λ is fixed at the truth, while β, ϕ, r, and κ are estimated.

Table 13: Oracle-lambda diagnostic: regression recovery.

rep	beta0_true	beta0_mean	beta0_cov	beta1_mean	beta1_cov	beta2_mean	beta2_cov
4	-2.280	-1.655	0	0.530	1	-0.484	1
7	-8.325	-1.299	0	0.497	1	-0.431	1

Table 14: Oracle-lambda diagnostic: latent recovery and guard counts.

rep	r_mean	phi_rmse	phi_cor	lambda_rmse	cor_loglam	kappa_guard
4	15.735	1.715	0.336	0.000	1.000	1652
7	17.027	19.884	0.143	0.000	1.000	465378

Fixing λ restores β_2 in both problem replicates: rep 4 has β_2 posterior mean about -0.484 and rep 7 has posterior mean about -0.431. Both cover the truth, -0.4. However, β_0, ϕ, and κ remain unstable, especially in rep 7. This confirms a β–λ ridge but leaves a remaining β_0–ϕ problem.

10.2 Step 2: Oracle lambda + oracle phi, estimate beta/r

In this diagnostic, both λ and ϕ are fixed at the truth. Only β, r, and κ are estimated.

Table 15: Oracle-lambda-plus-phi diagnostic: regression recovery.

rep	beta0_true	beta0_mean	beta0_cov	beta1_mean	beta1_cov	beta2_mean	beta2_cov
4	-2.280	-2.274	1	0.527	1	-0.356	1
7	-8.325	-8.329	1	0.499	1	-0.422	1

Table 16: Oracle-lambda-plus-phi diagnostic: latent checks and guard counts.

rep	r_mean	phi_rmse	phi_cor	lambda_rmse	cor_loglam	kappa_guard
4	16.724	4.16e-14	1.000	0.000	1.000	0
7	18.581	7.61e-13	1.000	0.000	1.000	0

This diagnostic is decisive. Once both λ and ϕ are fixed, all regression coefficients are recovered accurately and all numerical guards disappear. Therefore, the beta/r/kappa block itself is not the source of the S4B failure. The failure arises from joint estimation of regression, spatial, and temporal latent effects under low and heterogeneous exposure.

11 What low exposure does to the model

Low and heterogeneous exposure affects the model differently from global sparse-count reduction.

11.1 1. It creates local information imbalance

The exposure term is known, so low exposure is not itself an unknown parameter. However, low exposure reduces the amount of information available in the corresponding area-time cells. In low-exposure areas, observed counts are smaller and zeros are more common. These cells provide weaker information about local temporal risk.

11.2 2. It degrades low-exposure latent-risk recovery

The strongest stable-fit result is the low-versus-high exposure contrast. In stable and soft-warning fits, low-exposure areas have log-λ RMSE about 0.239, whereas high-exposure areas have log-λ RMSE about 0.081. This shows that exposure-poor areas recover latent temporal risk less accurately.

11.3 3. It leaves coefficient-specific fragility

The average β_2 posterior mean in stable-plus-soft-warning fits is close to the truth, but the between-replicate standard deviation is large. Thus, β_2 is not uniformly unrecoverable, but it remains fragile. This is consistent with the S4A findings: the β_2 design direction is easier to absorb into latent spatial-temporal effects than β_1.

11.4 4. It can still trigger posterior ridges in difficult replicates

Replicates 4 and 7 fail even though S4B is less globally sparse than S4A. This indicates that failure is not determined only by the overall zero proportion. It depends on the alignment among exposure, covariates, spatial effects, latent temporal paths, and realized counts.

In difficult replicates, the sampler can move along ridges such as

$$\beta_2 x_{2,tj} \leftrightarrow \log \lambda_{tj}$$

and

$$\beta_0 \leftrightarrow \phi_j.$$

These ridges preserve parts of the fitted intensity while producing unstable parameter values and numerical guard activity.

12 Practical implications

The S4B results suggest several practical implications for the MSSTNB model.

1. **Exposure heterogeneity should be treated as a stress regime.** Even when the overall count level is moderate, local low-exposure cells may contain too little information for reliable latent-risk recovery.
2. **Low- and high-exposure areas should be diagnosed separately.** Overall lambda RMSE can obscure the fact that recovery is much worse in exposure-poor areas.
3. **Fit-status classification remains necessary.** S4B has a 20% numerical-instability rate, so failure frequency is part of model performance.
4. **Oracle diagnostics are useful for distinguishing implementation errors from identifiability limits.** In S4B, oracle-$\lambda + \phi$ diagnostics restore beta/r recovery and eliminate guards.
5. **Real applications with strongly heterogeneous exposure may need stronger priors or aggregation.** Stabilizing strategies may include stronger priors on ϕ and λ, coarser spatial aggregation for very low-exposure areas, or reporting exposure-specific uncertainty and diagnostics.

13 Final conclusion

Scenario 4B confirms that low and heterogeneous exposure creates a distinct stress regime for the MSSTNB model. The average exposure is approximately 5% of the reference exposure, and the exposure distribution is strongly heterogeneous. The resulting data have average count about 9.56 and zero proportion about 0.125, which is less globally sparse than S4A. Nevertheless, the fitted model still shows 70% stable fits, 10% soft warnings, and 20% numerical instability.

Among stable and soft-warning fits, β_1 remains stable. β_2 is closer to the truth on average than in S4A, but remains highly variable across replicates. Latent temporal-risk recovery is substantially weaker than in the S3 control setting, and the degradation is concentrated in low-exposure areas. Oracle diagnostics show that fixing λ restores β_2, while fixing both λ and ϕ restores all regression effects and eliminates guards.

Therefore, S4B should be interpreted as a real exposure-stress result, not a generic implementation failure. Low and heterogeneous exposure weakens local information enough to degrade latent-risk recovery and, in difficult replicates, trigger the same regression–spatial–temporal posterior ridges observed in S4A.

14 Recommended reporting language

The following paragraph can be adapted for the manuscript or internal report:

> Scenario 4B evaluated the model under low and heterogeneous exposure by reducing the known exposure to approximately 5% of the reference exposure while introducing area-level exposure heterogeneity. The resulting data had an average count of approximately 9.6 and an average zero proportion of approximately 0.13. Under this regime, 70% of fitted replicates were stable, 10% produced a soft numerical warning, and 20% showed numerical instability. Among stable and soft-warning fits, β_1 remained accurately recoverable, whereas β_2 showed substantial between-replicate variability. Low-exposure areas had substantially worse latent temporal-risk recovery than high-exposure areas. Oracle diagnostics showed that fixing λ restored β_2, while fixing both λ and ϕ restored all regression effects and eliminated numerical guards. These results indicate that low and heterogeneous exposure creates a local information-imbalance stress regime: exposure-poor areas provide limited information for separating covariate effects, spatial effects, and latent temporal risk.